

2015 HKDSE Physics

Paper 1A Multiple Choice Questions

Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

An asterisk marks extension content in the original paper.

Question 1

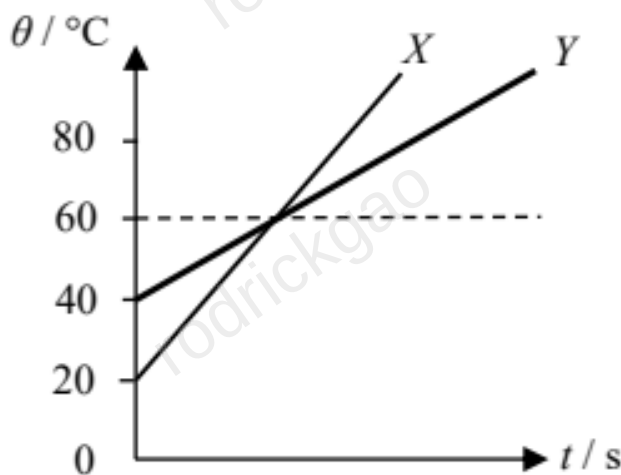
A driver turns off the engine after parking his car outdoors under the sun. Two hours later when getting into the car, he feels that the inside of the car is far hotter than outside. The best explanation is

- A. the car's engine is still generating heat after the engine has been switched off.
- B. the car's metal parts absorb infra-red radiation at a faster rate than the surroundings.
- C. the glass windows of the car trap infra-red radiation and a greenhouse effect results.**
- D. the surrounding air is a good insulator of heat which reduces heat loss by conduction.

Official answer: C

Sunlight enters through the glass and warms the interior. The warm interior emits infrared radiation, which the glass transmits poorly, reducing radiative heat loss. This is the greenhouse-effect explanation represented by C.

Question 2



Two objects X and Y are made of the same material. They are heated separately by heaters of the same power. The graph shows the variation of temperature θ of X and Y with time t . What is the ratio of mass of X to that of Y?

- A. 3 : 1
- B. 2 : 1
- C. 1 : 2**
- D. 2 : 3

Official answer: C

The heaters supply equal energy in the same time. With the same specific heat capacity, $m_X \Delta T_X = m_Y \Delta T_Y$. At the intersection, $\Delta T_X = 60 - 20 = 40^\circ\text{C}$ and $\Delta T_Y = 60 - 40 = 20^\circ\text{C}$. Thus $m_X/m_Y = 20/40 = 1/2$.

Question 3

When an object P is in contact with another object Q, heat flows from P to Q. P must have a higher

- (1) temperature.
- (2) internal energy.
- (3) specific heat capacity.

A. (1) only

B. (3) only

C. (1) and (2) only

D. (1) and (3) only

Official answer: A

Heat flows spontaneously from higher to lower temperature. Total internal energy also depends on mass and material, and specific heat capacity does not determine the direction of heat flow. Only (1) must be higher.

Question 4

A particle travels at 2.0 m s^{-1} due east for 1.5 s and then travels at 4.0 m s^{-1} due north for 1.0 s. What is the magnitude of its average velocity for the whole journey ?

A. 2.0 m s^{-1}

B. 2.8 m s^{-1}

C. 3.0 m s^{-1}

D. 5.0 m s^{-1}

Official answer: A

Displacement components are $2.0 \times 1.5 = 3.0 \text{ m}$ east and $4.0 \times 1.0 = 4.0 \text{ m}$ north. Magnitude of displacement = $\sqrt{(3^2 + 4^2)} = 5.0 \text{ m}$. Hence |average velocity| = $5.0/(1.5 + 1.0) = 2.0 \text{ m s}^{-1}$. Do not use total distance.

Question 5

A constant net force acting on an object of mass m_1 produces an acceleration a_1 while the same force acting on another object of mass m_2 produces an acceleration a_2 . If this net force acts on an object of mass $(m_1 + m_2)$, what would be the acceleration produced?

A. $a_1 + a_2$

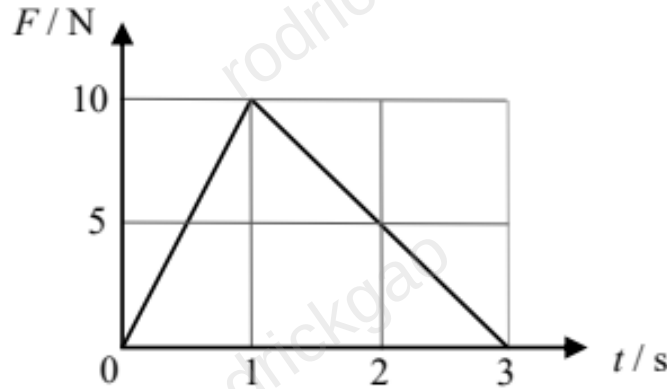
B. $(a_1 + a_2)/2$

C. $a_1 a_2 / (a_1 + a_2)$

D. $2a_1 a_2 / (a_1 + a_2)$

Official answer: C

Newton's second law gives $m_1 = F/a_1$ and $m_2 = F/a_2$. Therefore $a = F/(m_1 + m_2) = 1/(1/a_1 + 1/a_2) = a_1 a_2 / (a_1 + a_2)$.

Question 6

An object of mass 3 kg is initially at rest on a smooth horizontal ground. A force F is applied horizontally to the object such that the magnitude of F varies with time t as shown. What is the speed of the object at $t = 3$ s? Neglect air resistance.

A. 2.5 m s^{-1}

B. 5 m s^{-1}

C. 10 m s^{-1}

D. 15 m s^{-1}

Official answer: B

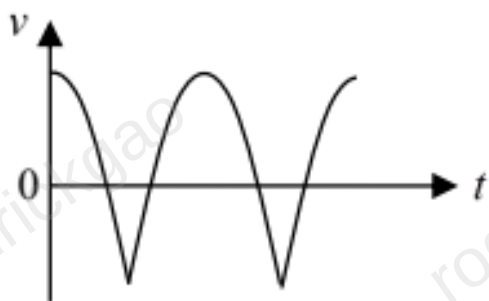
Impulse equals the area under the force–time graph. The graph is a triangle of base 3 s and height 10 N, so $\Delta p = \frac{1}{2} \times 3 \times 10 = 15 \text{ N s}$. Initially at rest, the 3 kg object reaches $v = \Delta p/m = 15/3 = 5 \text{ m s}^{-1}$.

Question 7

A rubber ball bounces vertically up and down from the ground. Which graph best shows the variation of its velocity v with time t if the collisions are elastic? Neglect air resistance.

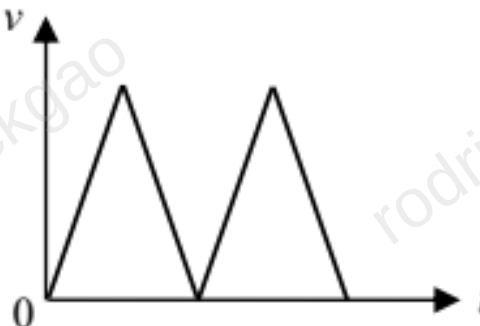
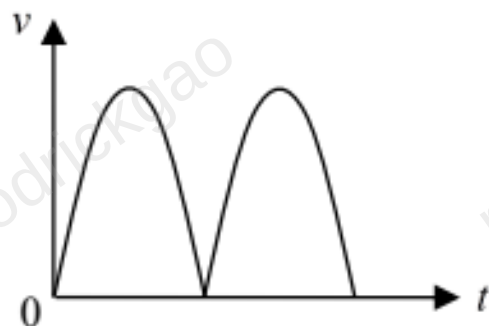
A.

B.



C.

D.

**Official answer: B**

Take upward as positive. Between bounces, acceleration is constant at $-g$, so velocity decreases linearly with time. Each elastic bounce reverses the velocity without changing its magnitude, giving a sudden jump from negative to positive velocity. This is graph B.

Question 8

An object falls from P to Q as shown. Throughout the motion, air resistance increases with the speed of the object. Which of the following descriptions is/are correct?

- (1) The net force acting on the object is constant throughout the motion.
- (2) The magnitude of the object's acceleration decreases from P to Q.
- (3) The kinetic energy gained by the object from P to Q is equal to its loss in gravitational potential energy.

A. (1) only

B. (2) only

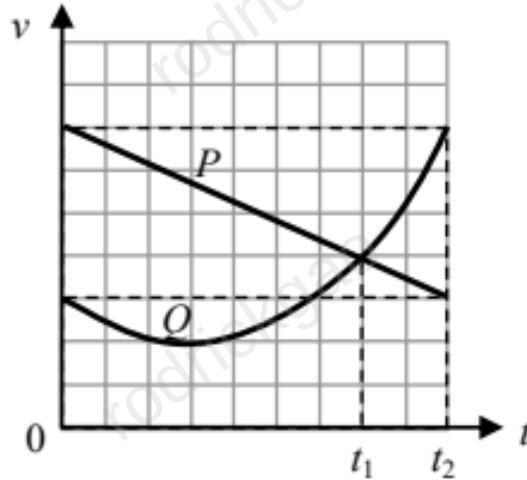
C. (1) and (3) only

D. (2) and (3) only

Official answer: B

As speed rises, upward air resistance increases, so the downward net force $mg - F_{\text{air}}$ and acceleration decrease. Some gravitational potential energy becomes internal energy through air resistance, so the kinetic-energy gain is smaller than the potential-energy loss. Only (2) is correct.

Question 9



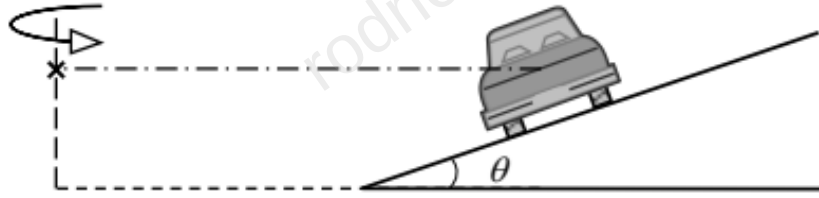
The figure shows the velocity-time (v - t) graph of two cars P and Q travelling along the same straight road. At $t = 0$, the cars are at the same position. Which deductions about the cars between $t = 0$ and $t = t_2$ are correct?

- (1) P and Q are always travelling in the same direction.
- (2) At $t = t_1$, the separation between P and Q is at a maximum.
- (3) At $t = t_2$, Q lags behind P.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: D

- (1) Both velocity curves remain positive, so both cars move in the same direction.
- (2) Separation grows while $v_P > v_Q$ and decreases after the curves cross. It is therefore greatest at t_1 .
- (3) By t_2 , the area under P's curve is still greater than that under Q's curve, so Q has not caught up. All three statements are correct.

Question 10 *

The figure shows the rear view of a car of mass m which travels along a circular road banked with an angle θ to the horizontal. The car moves at a certain speed such that it experiences no frictional force along the inclined surface. Which of the following represents the centripetal force on the car?

- A. $mg \sin \theta$
- B. $mg \sin \theta \cos \theta$
- C. $mg \cos \theta / \sin \theta$
- D. $mg \sin \theta / \cos \theta$

Official answer: D

With no friction, the forces are weight mg and the normal reaction N . Vertical balance gives $N \cos \theta = mg$. The horizontal component supplies the centripetal force: $F_c = N \sin \theta = mg \sin \theta / \cos \theta$.

Question 11

The gravitational force exerted on the Earth by the Sun is F_0 . The gravitational force exerted on the Sun by the Earth is

- A. equal to F_0 and in the same direction.
- B. equal to F_0 and in the opposite direction.
- C. much smaller than F_0 and in the same direction.
- D. much smaller than F_0 and in the opposite direction.

Official answer: B

The two gravitational forces form a Newton's third-law pair. They act on different bodies and are equal in magnitude but opposite in direction, irrespective of the bodies' unequal masses.

Question 12

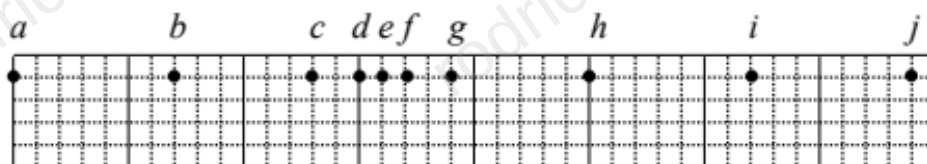


Figure (a)

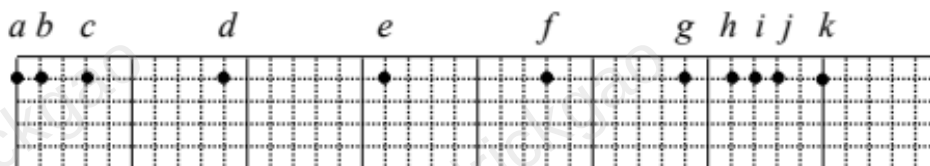


Figure (b)



Figure (c)

A series of particles is uniformly distributed along a slinky spring initially. When a travelling wave propagates along the slinky spring from left to right, Figure (a) shows the positions of the particles at a certain instant. Figures (b) and (c) respectively show their positions 0.05 s and 0.1 s later. Which of the following is/are a possible frequency of the wave?

- (1) 10 Hz
- (2) 20 Hz
- (3) 40 Hz

A. (1) only

B. (2) only

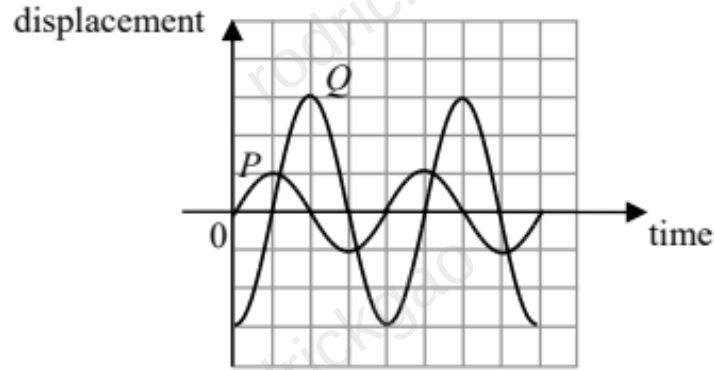
C. (3) only

D. (1), (2) and (3)

Official answer: A

The pattern after 0.05 s differs from the initial pattern, while that after 0.10 s repeats it. A frequency of 10 Hz gives $T = 0.10$ s and the different half-period pattern at 0.05 s. At 20 or 40 Hz, 0.05 s would be an integer number of periods and the pattern would already repeat. Only 10 Hz fits.

Question 13



Two waves P and Q travel in the same direction and meet at a point. The above graph shows the variation of the displacement of each wave with time at that point. Which of the following statements is/are correct?

- (1) P and Q have the same frequency.
- (2) The oscillation due to P is in anti-phase with that due to Q.
- (3) The amplitude of the resultant wave at that point is four times the amplitude of P.

A. (1) only

B. (3) only

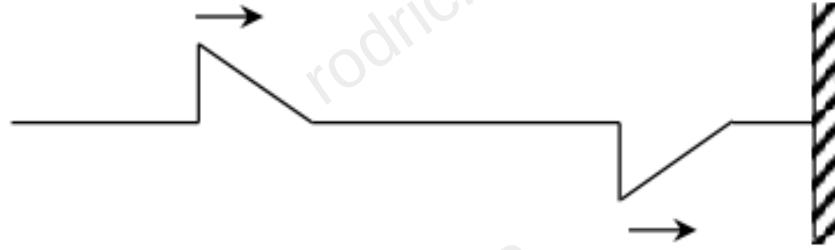
C. (1) and (2) only

D. (2) and (3) only

Official answer: A

The two waves have equal periods, hence equal frequencies. Their phase difference is a quarter-cycle, not half a cycle. With $A_Q = 3A_P$, the resultant amplitude is $\sqrt{A_P^2 + A_Q^2} = \sqrt{10} A_P$, not $4A_P$. Only (1) is correct.

Question 14



Two pulses of the same shape travel along a stretched string with one end fixed to the wall as shown above. Which of the following can be the resultant waveform at different instants later?

(1)



(2)



(3)



A. (1) only

B. (3) only

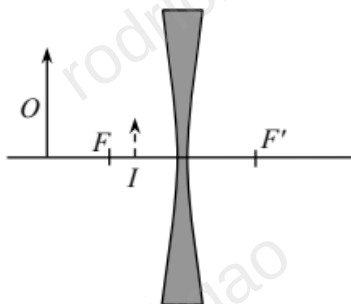
C. (1) and (2) only

D. (2) and (3) only

Official answer: D

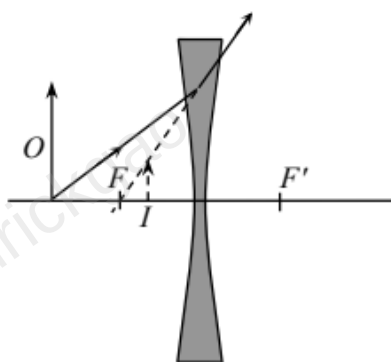
Reflection at the fixed end inverts a pulse and reverses its direction of travel. The leading downward pulse therefore returns as an upward pulse with its profile reversed. Before overlap the two upward pulses can form (2); when fully superposed their complementary slopes add to the rectangular pulse (3). They do not cancel to a flat string. Hence (2) and (3) are possible.

Question 15

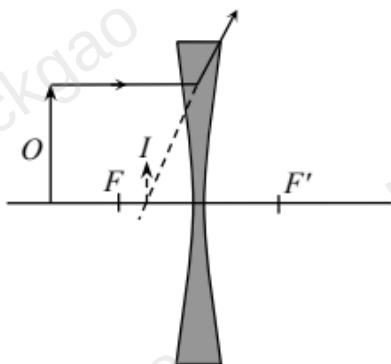


An object O placed in front of a concave lens forms an image I as shown. F and F' are the foci of the lens.

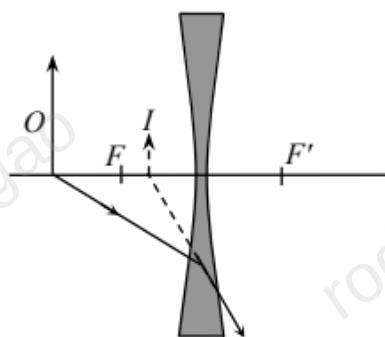
A.



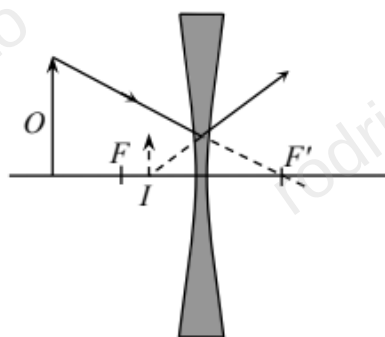
C.



B.



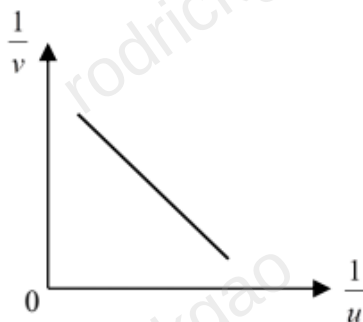
D.



Official answer: B

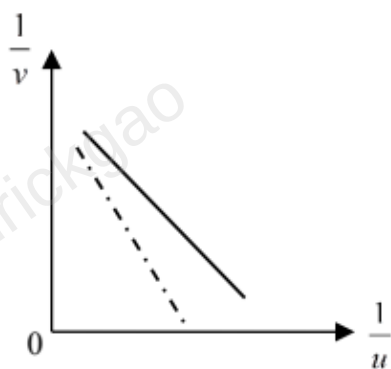
For a diverging lens, a refracted ray diverges and its backward extension passes through the corresponding virtual image point. In B the ray from the object's base is bent away from the principal axis and its backward extension passes through the image's base. A ray aimed at the far focus would emerge parallel, not follow D.

Question 16 *

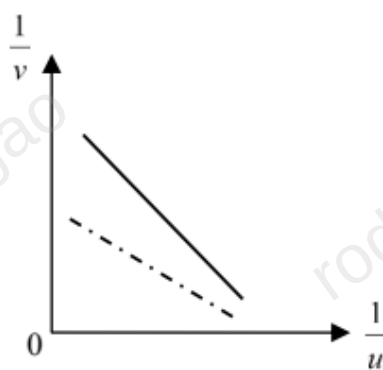


A student uses a convex lens to investigate the variation of image distance v with object distance u for real images. The graph of $1/v$ plotted against $1/u$ is shown above. If a convex lens of longer focal length is used, what would be the expected result (in dotted lines)?

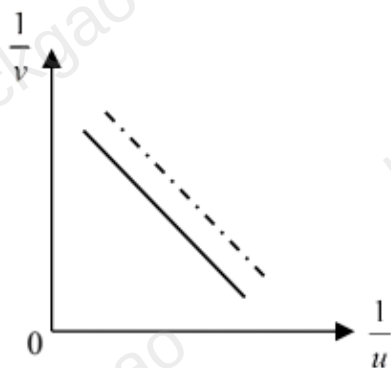
A.



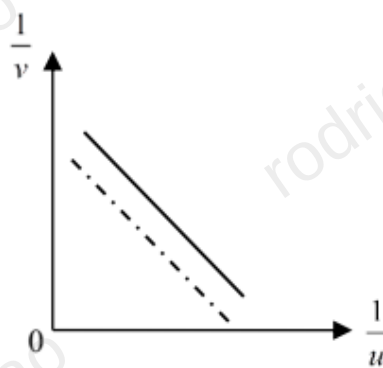
B.



C.



D.

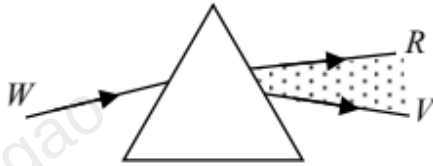
**Official answer: D**

The lens equation gives $1/v = -1/u + 1/f$. The slope is always -1 . Increasing f decreases $1/f$, so the new line is parallel to and below the original line: D.

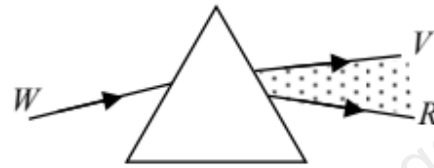
Question 17

Which diagrams below correctly show the spectra formed from white light by a glass prism and a diffraction grating respectively? It is known that red light travels faster than violet light in glass. (R = red, V = violet, W = white)

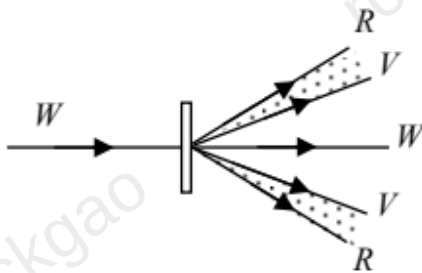
(1)



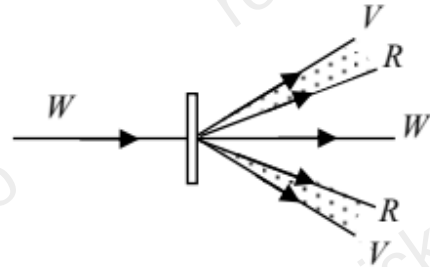
(2)



(3)



(4)



A. (1) and (3) only

B. (1) and (4) only

C. (2) and (3) only

D. (2) and (4) only

Official answer: A

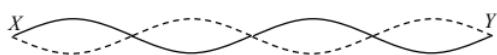
In glass, red travels faster and has the smaller refractive index, so it is deviated less than violet: (1). For a grating, $d \sin \theta = n\lambda$; red has the larger wavelength and therefore the larger diffraction angle: (3).

Question 18



A string is set to vibrate at frequency f such that a standing wave is formed between two fixed supports X and Y as shown. If the tension in the string is increased by adding weight gradually while the frequency is kept at f , which of the following is a possible mode of vibration at a steady state?

A.



B.



C.



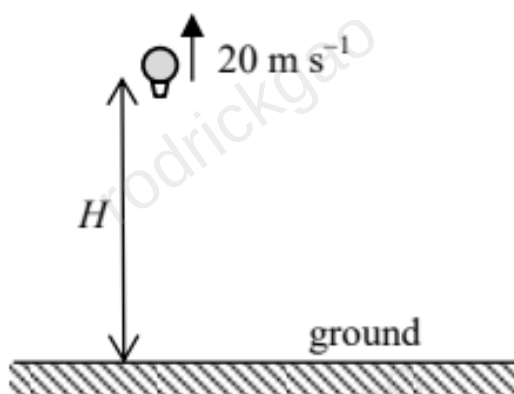
D.



Official answer: D

Increasing tension increases the wave speed. At fixed frequency, $\lambda = v/f$ therefore increases. With fixed separation XY, the number of half-wavelength loops decreases from two to one. The possible new mode is D.

Question 19



A balloon is rising at a uniform speed of 20 m s^{-1} . When the balloon is at an altitude H as shown, it sends a sound signal towards the ground. After 5 s, the balloon receives the echo of the signal. Estimate H . Given: speed of sound in air = 340 m s^{-1} .

A. 1600 m

B. 850 m

C. 800 m

D. 750 m

Official answer: C

In 5 s the balloon rises $20 \times 5 = 100 \text{ m}$. The sound travels H downward and $H + 100$ upward, so $340 \times 5 = 2H + 100$. Hence $H = 800 \text{ m}$.

Question 20

Which of the following gives the order of magnitude of the wavelengths of ultra-violet radiation and microwave in a vacuum?

- A. Ultraviolet: 10^{-8} m; microwave: 10^{-2} m
- B. Ultraviolet: 10^{-8} m; microwave: 10^{-5} m
- C. Ultraviolet: 10^{-10} m; microwave: 10^{-2} m
- D. Ultraviolet: 10^{-10} m; microwave: 10^{-5} m

Official answer: A

Ultraviolet wavelengths are shorter than visible light, with 10^{-8} m a representative order. Microwaves can have centimetre wavelengths, of order 10^{-2} m. The matching pair is A.

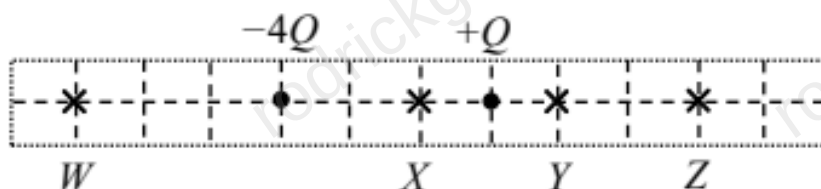
Question 21

Three conducting balls are suspended by insulating threads. Any two of them are found to attract each other if brought close to each other. Which conclusion below is correct?

- A. Only one ball is uncharged while the other two carry like charges.
- B. Only one ball is uncharged while the other two carry unlike charges.
- C. Only one ball is charged.
- D. All three balls are charged.

Official answer: B

A positively charged ball and a negatively charged ball attract each other. Either also attracts a neutral conducting ball through charge redistribution. Three charged balls cannot all have opposite signs pairwise, and two neutral balls do not supply the required mutual electrostatic attraction. Thus one is neutral and the other two carry unlike charges.

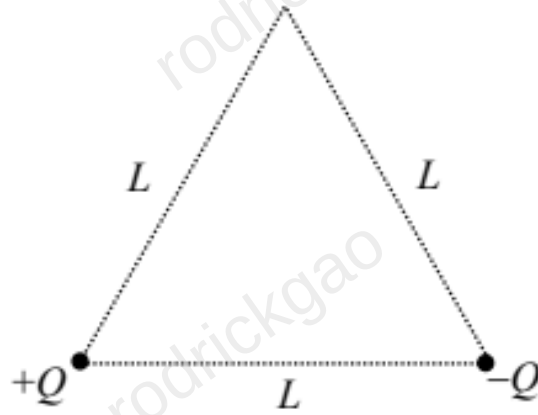
Question 22

Two point charges $-4Q$ and $+Q$ are fixed as shown. At which point indicated in the figure is the resultant electric field due to these two charges zero?

- A. W
- B. X
- C. Y
- D. Z

Official answer: D

Between the unlike charges, both fields point toward $-4Q$ and cannot cancel. Outside, cancellation must lie on the $+Q$ side, nearer the smaller charge. Equal magnitudes require $4/r_-^2 = 1/r_+^2$, hence $r_- = 2r_+$. Point Z satisfies this distance ratio.

Question 23 *

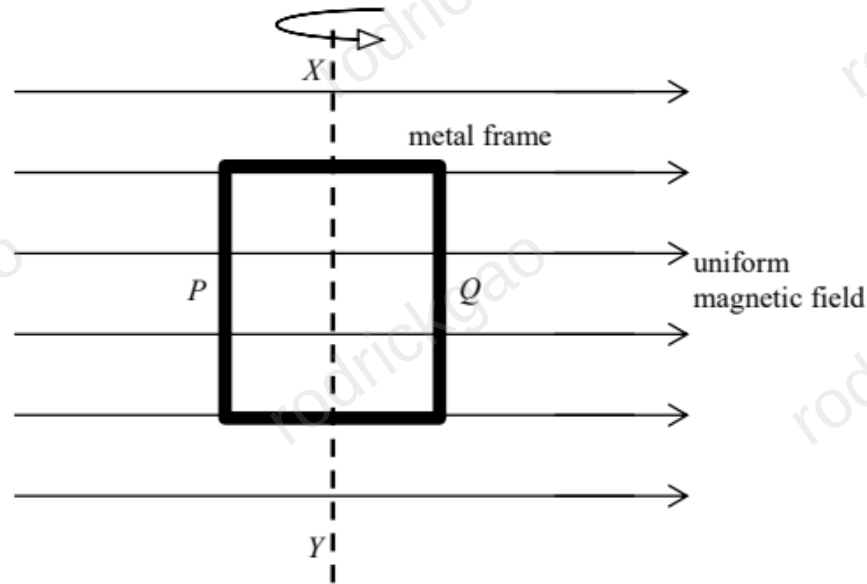
Point charges $+Q$ and $-Q$ are fixed respectively at two vertices of an equilateral triangle with sides of length L as shown. What is the minimum energy required for bringing another point charge $+Q$ from infinity to the third vertex?

- A. 0
- B. $(1/(4\pi\epsilon_0))(Q^2/L)$
- C. $(1/(4\pi\epsilon_0))(2Q^2/L)$
- D. $(1/(4\pi\epsilon_0))(3Q^2/L)$

Official answer: A

At the third vertex, the electric potentials from $+Q$ and $-Q$ cancel: $V = (1/(4\pi\epsilon_0))(Q/L - Q/L) = 0$. Quasistatic external work from infinity is $\Delta U = QV = 0$.

Question 24



A rectangular metal frame is made to rotate steadily about its axis XY in a uniform magnetic field. At the instant shown, the frame is in the plane of the paper and side P is moving out of the paper while side Q is moving into the paper. Which statement is INCORRECT at this instant?

- A. The induced e.m.f. in the frame is at a maximum.
- B. The induced current produced in the frame is flowing in anti-clockwise direction.
- C. The magnetic force acting on side P is in a direction pointing into the paper.
- D. The magnetic forces acting on the frame produce a moment opposing the frame's rotation.

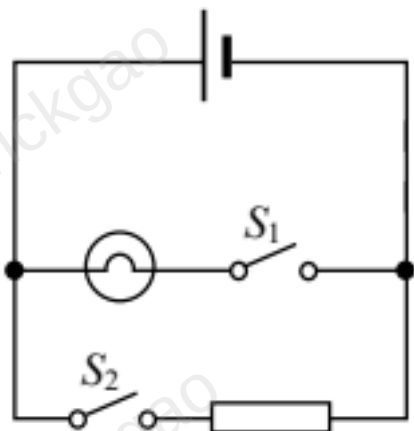
Official answer: B

At this instant the field lies in the frame's plane, so the flux is zero but its rate of change is greatest. For side P moving out of the page in a rightward field, the generator rule gives upward conventional current in P; the loop current is clockwise. B is therefore incorrect. The magnetic force opposes the motion by Lenz's law.

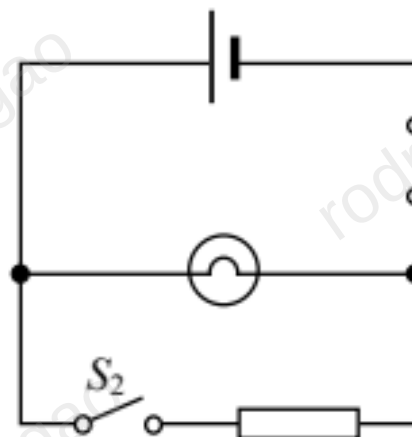
Question 25

For safety purposes, the driver seat of a car is equipped with a seat belt warning light. When the driver seat is occupied, the switch S_1 under his seat will close. If the seat belt is not yet fastened, switch S_2 will remain open and the warning light will light up. If the seat belt is fastened, the switch S_2 will close and the warning light will shut off. Which circuit design below is the best design?

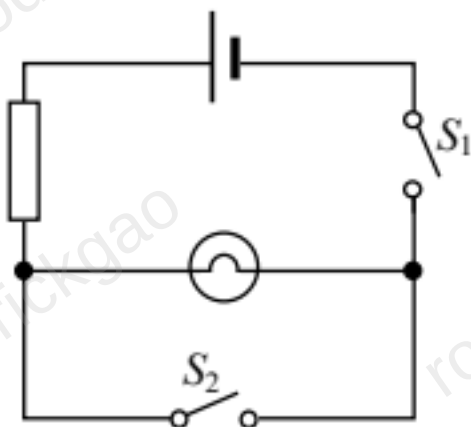
A.



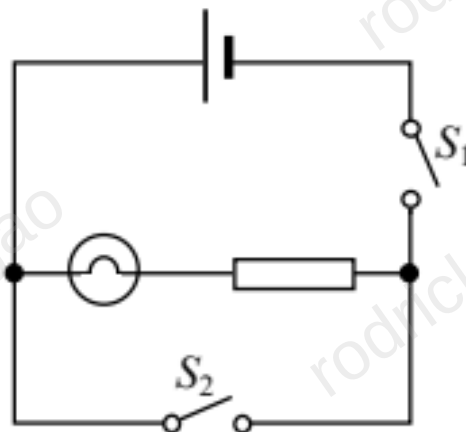
B.



C.

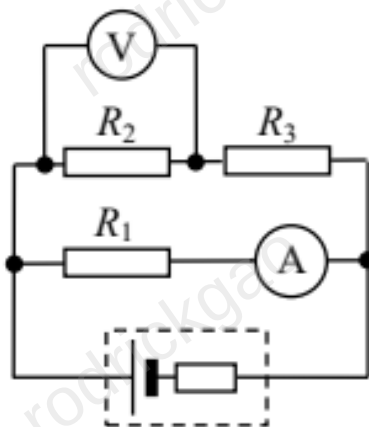


D.

**Official answer: C**

In C, S_1 is in series with the supply, so an unoccupied seat gives no current. With S_1 closed and S_2 open, current passes through the lamp. Closing S_2 bypasses the lamp and turns it off, while the series resistor still limits current. D would short the supply when both switches close.

Question 26



In the circuit shown, the cell has a finite internal resistance and both meters are ideal. In which situation below will the readings of the ammeter and the voltmeter suddenly increase?

- A. R_1 is faulty and becomes a short circuit.
- B. R_2 is faulty and becomes a short circuit.
- C. R_3 is faulty and becomes a short circuit.
- D. R_2 is faulty and becomes an open circuit.

Official answer: D

When R_2 becomes open circuit, the upper branch draws no current. Total current from the cell decreases, so the internal voltage drop decreases and terminal voltage rises. The ammeter current through unchanged R_1 rises. There is no drop across R_3 , so the voltmeter now reads the full, increased terminal voltage. Both readings increase.

Question 27 *

A copper wire of uniform cross-section carries a current of 0.5 A. Each metre of the wire contains 10^{22} free electrons. Find the magnitude of the drift velocity of the electrons in the wire.

- A. $2.5 \times 10^{-3} \text{ m s}^{-1}$
- B. $7.8 \times 10^{-3} \text{ m s}^{-1}$
- C. $3.1 \times 10^{-4} \text{ m s}^{-1}$
- D. $9.7 \times 10^{-4} \text{ m s}^{-1}$

Official answer: C

If the drift speed is v , the electrons in a length v metres pass a cross-section each second. Thus $I = (10^{22} \text{ m}^{-1})e v$. Therefore $v = 0.5 / (10^{22} \times 1.60 \times 10^{-19}) = 3.125 \times 10^{-4} \text{ m s}^{-1} \approx 3.1 \times 10^{-4} \text{ m s}^{-1}$.

Question 28

Which statement is NOT a reason why mains sockets at home are connected in parallel instead of a series circuit ?

- A. Electrical appliances connected to different sockets can be switched on or off independently.
- B. Voltage supplied to each socket is fixed and all electrical appliances can operate at their rated voltages.
- C. The current supplied can be reduced and thinner cables can then be used.
- D. When an electrical appliance breaks down and becomes an open circuit, other appliances can still work normally.

Official answer: C

Parallel branches receive the same supply voltage and can operate independently. Their currents add in the supply cable; parallel connection does not reduce total current or justify thinner supply cables. Therefore C is not a reason.

Question 29

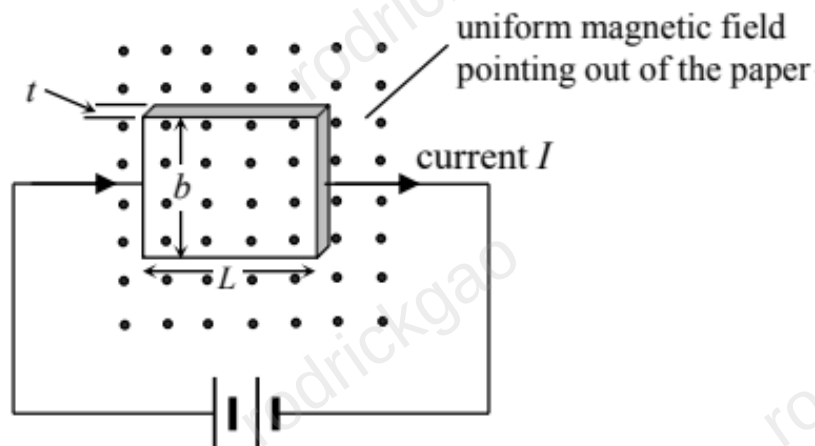
An electric iron of 1800 W sold in Hong Kong (220 V 50 Hz) is connected to a 110 V 60 Hz mains socket in another country. How does its performance compare on the same ironing setting ?

- A. The electric iron does not work because the a.c. supply is 60 Hz instead of 50 Hz.
- B. The electric iron is as hot as when it is used in Hong Kong.
- C. The electric iron is hotter than when it is used in Hong Kong.
- D. The electric iron is colder than when it is used in Hong Kong.

Official answer: D

For the resistive heating element, $P = V^2/R$. At approximately unchanged resistance, halving the voltage reduces heating power to one quarter: $1800/4 = 450$ W. The iron heats less strongly and is colder on the stated setting; the change from 50 to 60 Hz does not prevent resistive heating.

Question 30 *



A slice of conductor (with a certain resistivity) of thickness t , breadth b and length L is connected to a battery of constant e.m.f. and negligible internal resistance. A steady current I passes through it as shown. When a uniform magnetic field is applied normally to the slice, a Hall voltage V is set up between two of its opposite faces. If the thickness and the breadth of the slice are reduced to $1/2 t$ and $1/2 b$ respectively, what would the following quantities be?

- A. Current: $I/4$; Hall voltage: $V/4$
- B. Current: I ; Hall voltage: $V/4$
- C. Current: $I/4$; Hall voltage: $V/2$
- D. Current: I ; Hall voltage: $V/2$

Official answer: C

The current cross-sectional area is bt . Halving both b and t reduces it to one quarter, so $R = \rho L/(bt)$ becomes $4R$ and the current becomes $I/4$ at fixed battery voltage. Hall voltage is proportional to I/t , so its new value is $(1/4)/(1/2) = 1/2$ of the original: $V/2$.

Question 31

Which of the following nuclear reactions is/are spontaneous reaction(s)?

- (1) ${}_{11}^{24}\text{Na} \rightarrow {}_{12}^{24}\text{Mg} + {}_{-1}^0\text{e}$
- (2) ${}_{5}^{10}\text{B} + {}_0^1\text{n} \rightarrow {}_3^7\text{Li} + {}_2^4\text{He}$
- (3) ${}_1^2\text{H} + {}_1^3\text{H} \rightarrow {}_2^4\text{He} + {}_0^1\text{n}$

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: A

(1) is spontaneous beta decay of an unstable nucleus. Reaction (2) requires an incident neutron, and (3) requires nuclei to be brought sufficiently close for fusion. These are induced reactions in this context. Only (1) is spontaneous.

Question 32

Workers in nuclear power plants wear clothes with film badges to measure the dosage of radiation received over a period of time. Which type of radiation below CANNOT be monitored by the film badges?

- A. α -radiation
- B. β -radiation
- C. γ -radiation
- D. X-rays

Official answer: A

Alpha radiation is stopped by the protective covering of an ordinary film badge and cannot reach the photographic film. Beta radiation, gamma radiation and X-rays can expose the film. The answer is A.

Question 33 *

A piece of wood found in an archaeological site is dated using carbon-14 dating method. It registers a corrected count rate of 11.0 counts per minute while a fresh wood sample cut from the same kind of trees gives a corrected count rate of 15.6 counts per minute. What is the approximate age of the wood found in the archaeological site? Given: half-life of carbon-14 is 5730 years.

- A. 890 years
- B. 1300 years
- C. 2000 years
- D. 2900 years

Official answer: D

Use count rate proportional to activity for the compared wood samples: $11.0/15.6 = (1/2)^{t/5730}$. Thus $t = 5730 \ln(15.6/11.0)/\ln 2 \approx 2.89 \times 10^3$ years, closest to 2900 years.